MODULES OVER A LOCALISATION ARE EXACTLY THE MODULES ON WHICH EVERY ELEMENT OF S ACTS INVERTIBLY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted proof ends with the author's sentence that the verification of the two functors being quasi-inverse is omitted; that sentence is kept verbatim inside the span and both functors are described in full.

Editorial note. The counted spans contain no \ref and no \cite, so the card carries no navigation stub and no bibliography.

Allowed standard prerequisites (editorial).

- Restriction of scalars along the localisation map. Used in the first half of the counted proof (lines 1346-1348). Turns an S^{-1}R-module into an R-module on which every s in S acts invertibly.
- Making an R-module an S^{-1}R-module from invertible actions. Used in the second half of the counted proof (lines 1348-1350). Defines the action of x/s as multiplication by x followed by the inverse of the action of s.

Lemma 0.1. *Let R be a ring. Let $S \subset R$ be a multiplicative subset. The category of $S^{-1}R$-modules is equivalent to the category of R-modules N with the property that every $s \in S$ acts as an automorphism on N.*

Proof. The functor which defines the equivalence associates to an $S^{-1}R$-module M the same module but now viewed as an R-module via the localization map $R \to S^{-1}R$. Conversely, if N is an R-module, such that every $s \in S$ acts via an automorphism s_N, then we can think of N as an $S^{-1}R$-module by letting x/s act

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via $x_N \circ s_N^{-1}$. We omit the verification that these two functors are quasi-inverse to each other. $\square$